

AI Tutor: A Retrieval-Augmented Q&A System for Textbook Content

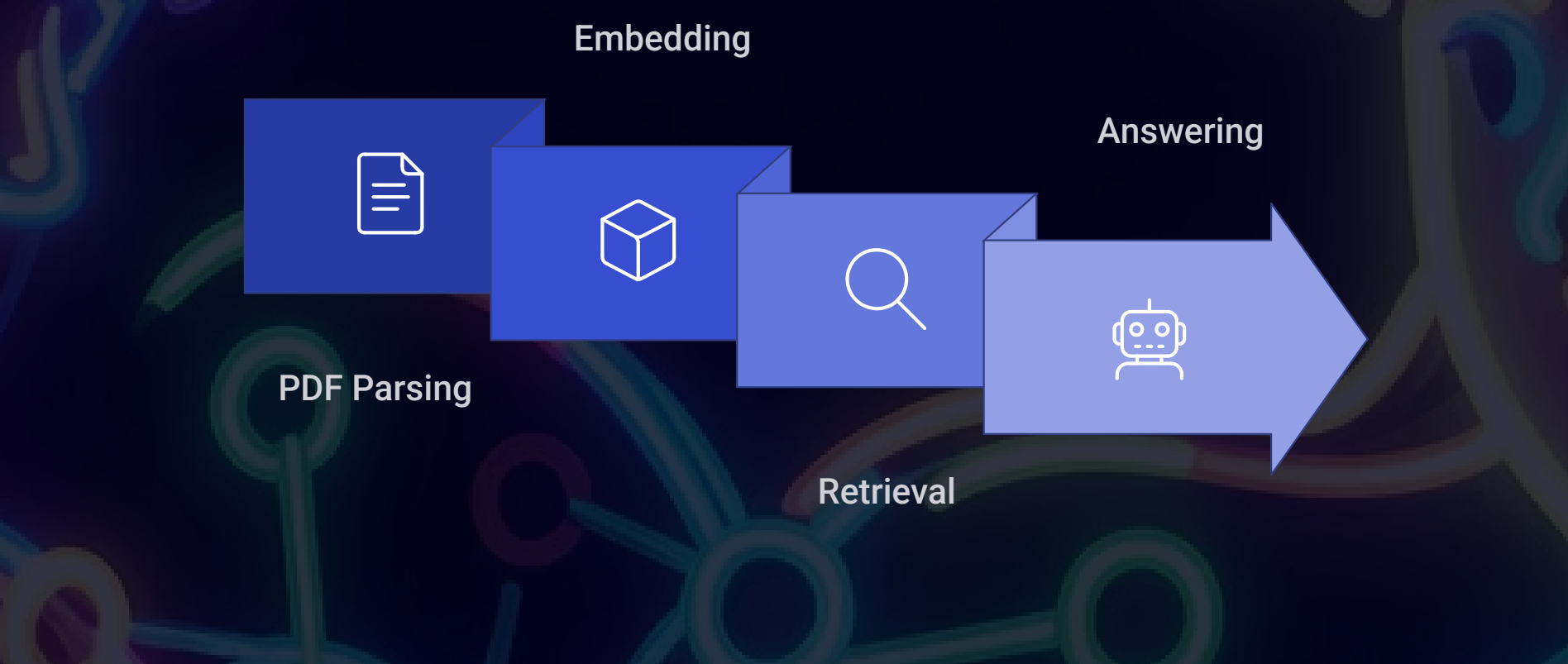
A RAG-based system that reads Class 11–12 NCERT science textbooks and answers student questions — combining semantic search with Llama-2 for accurate, hallucination-free responses.

Purpose
Instantly retrieve concepts from dense textbooks, saving time and ensuring no key detail is missed.

Applications
Educational chatbots, homework helpers, and teacher aides focused on specific curricula.

Core Approach
Fetches exact textbook passages, then uses Llama-2 to compose natural answers — avoiding hallucinations.

Project Architecture



Heavy lifting (index building) is done once offline; real-time queries involve quick vector searches and one LLM call.

Key Technologies

Technology	Role	Analogy
PyMuPDF	Extract text from PDFs	Smart scanner for textbooks
Sentence-Transformers	Generate 384-dim embeddings	Sentences → numeric fingerprints
FAISS	Fast vector similarity search	GPS for vector addresses
Llama-CPP / Llama-2	Generate answers from context	On-device AI tutor brain
NLTK & Rouge-Score	BLEU/ROUGE evaluation	Teacher grading system

Code Highlights

parse_pdf
Maps filename prefixes to subject, class, and chapter metadata.

chunk_text
Splits text into 200-word overlapping chunks (50-word overlap) for context continuity.

retrieve
Encodes the question, searches FAISS, returns top-K chunks with similarity scores.

generate_answer
Builds a labeled context prompt; Llama-2 (temp=0.2, 2024-token window) generates the answer.

Core AI Concepts

Transformers & Embeddings
Transformers use attention to capture context. Embeddings map text to vectors — similar sentences land closer in vector space.

RAG & Vector Search
Retrieves relevant textbook passages via FAISS (cosine/Euclidean similarity), then grounds Llama-2's answer in real content.

Prompt Engineering
System instruction restricts Llama-2 to provided context and directs it to admit when an answer isn't found.

Challenges & Future Directions

Current Challenges

- Noisy PDF extraction; complex layouts may need OCR
- Chunk size tuning (200 words, 50-word overlap)
- Context window limits (2024 tokens); truncation to 600 chars per chunk
- Hallucinations mitigated by explicit system instructions

Future Improvements

- Web UI (Streamlit/Flask) for browser-based access
- Incremental corpus updates without full rebuild
- Fine-tune Llama-2 on NCERT Q&A pairs
- Hybrid dense-sparse retrieval for better coverage
- Upgrade to Llama-3 with larger context windows

Impact: Empowers students with instant, reliable textbook-grounded explanations — demonstrating how AI can personalize education at scale.